DERANGULA SUSHANTH

AI/ML ENGINEER — GENERATIVE AI, LLM APPLICATIONS & GRAPH-READY DATA SCIENCE

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building Generative AI and LLM-powered applications — RAG pipelines, embeddings, text chunking, and prompt engineering — alongside deep learning model development (CNNs, TensorFlow, PyTorch) for predictive analytics, computer vision, and NLP. Skilled at translating business requirements into deployable ML/DL and GenAI systems, with production experience across Python, FastAPI, and REST APIs. Has applied ML-based decision scoring and rule-based logic to real-world risk and document-intelligence problems (OCR-based document processing, unstructured data, predictive risk/denial scoring), and is extending this into vector database-backed retrieval (FAISS, Chroma), agentic AI workflows, and cloud-native AI deployment (AWS, Azure OpenAI, Azure AI Foundry). Strong in feature engineering, hyperparameter tuning, model compression, and model evaluation, with a track record of owning the full lifecycle from requirement translation through deployment.

TECHNICAL SKILLS

Generative AI & LLMs: LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Text Chunking, Conversational AI Workflows, IBM watsonx.ai, Granite LLM

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, TensorFlow, PyTorch, scikit-learn, CNNs, Feature Engineering, Hyperparameter Tuning, Model Compression, Model Evaluation, Model Deployment,XGBoost/LightGBM,Random Forest,Precision,ROC-AUC

Risk, Document & Computer Vision Intelligence: Predictive Risk/Decision Scoring, Rule-Based Validation Logic, OCR-Based Document Processing, Metadata Extraction, OpenCV, Image-Based Feature Extraction, Object Detection Concepts, Structured & Unstructured Data, Pandas, NumPy

APIs, Programming & Tools: Python, FastAPI, Flask, RESTful APIs, SQL, JavaScript (ReactJS), Git, Docker (basic), DevOps Fundamentals, Google Colab, Kaggle, VS Code, Jupyter, Streamlit, Vercel, Matplotlib

Currently Building Toward: Vector Databases (FAISS, Chroma), Agentic AI Frameworks & MCP, Cloud AI Deployment (AWS, Azure OpenAI, Cognitive Services, Azure AI Foundry)

WORK EXPERIENCE

Data Science Intern | Vajra.ai | Hyderabad, India *Jan 2026 – Jul 2026*

- **Requirement-to-model translation:** Partnered with the AI R&D team to convert business and product requirements into ML/DL model specifications for predictive analytics, computer vision, and NLP use cases.
- **Pipeline automation & API integration:** Owned pipeline automation end-to-end — from requirement definition through Python and RESTful API integration — supporting reliable, production-oriented model deployment workflows.
- **Model optimization:** Improved model performance and reliability through structured experimentation, including hyperparameter tuning, feature engineering, and model compression.
- **Documentation & collaboration:** Documented experiments and workflow decisions, maintained version-controlled repositories in Git, and communicated results to cross-functional technical teams.

PROJECTS

Interview Trainer Agent

Python, FastAPI, ReactJS, IBM watsonx.ai, Granite LLM, REST APIs, RAG, Embeddings, Prompt Engineering | github.com/sushanth0963/Interview_Trainer_Agent

- **Generative AI application:** Built an end-to-end Generative AI application using IBM watsonx.ai and Granite LLM, applying prompt engineering to generate personalized, role-specific interview questions and model answers.
- **Full-stack AI service:** Built and integrated a full-stack solution (ReactJS frontend + FastAPI backend) exposing LLM functionality via scalable REST APIs for real-time user interaction and adoption testing.
- **RAG pipeline design:** Designed a Retrieval-Augmented Generation workflow — including text chunking and embedding-based retrieval — to deliver accurate, contextual LLM responses across multiple domains and experience levels.

Healthcare Claim Denial Predictor (Risk & Document Intelligence)

Python, Machine Learning, OCR, Rule-Based Logic, Streamlit | github.com/sushanth0963/Health-Care-Claim-Denial-Predictor

- **Document automation & OCR:** Built an end-to-end workflow processing unstructured document data via OCR, combined with rule-based business logic and ML-based decision scoring, to predict insurance claim denial probability — an applied risk-scoring problem.
- **Stakeholder validation dashboard:** Built an interactive Streamlit dashboard enabling scenario testing and transparent, explainable rule- and model-based insights for non-technical stakeholders.
- **Business impact:** Delivered measurable impact by design — faster claim review turnaround and improved decision-making efficiency.

EDUCATION

Bachelor of Technology, Information Technology — CMR Engineering College | 2021 – 2025

CERTIFICATIONS & TRAINING

Data Analytics & AI — Social Prachar, Hyderabad, India | Aug 2025 – May 2026

- Industry-focused training in Python, SQL, Machine Learning, Deep Learning, and NLP, with real-time predictive modeling projects.